



Thapar Institute of Engineering & Technology

(Deemed to be University)

PATIALA

298971

Answer Sheet No.

Name Rudraresh Garg Roll No. 1024030913 Class/Group 2C64Course No. UCS301 Course Name Data StructuresName of the instructor Dr. Ritesh Date 29th Sept, 2015

INSTRUCTIONS

1. Write on both sides of the answer-book.
2. Do not remove or tear off any page from this answer book.
3. Candidates found guilty of unfair means or misconduct shall be dealt with as per Institute rules on unfair means.
4. Write your Roll No. on the Question Paper.
5. Write Page No. of each question (which you have attempted) in the table given below.

Evaluation:

Signature of the Invigilator

Q.No.	I	II	III	IV	V	VI	VII	VIII	IX	X	Total Marks
Page No.	4	5		1/2							
Marks	3+1	NA	0+4	0+0							

Total Marks in words:

Signature of the Examiner

Start Writing from here :-

Q: 1(a) • function 1: (void func1)

for (int i = 1; i ≤ n, i++) { → goes n times

for (int j = 1, j ≤ n, j++) { → goes n times

for (int k = 1, k ≤ 10, k++) { → goes 10 times

count++ { } }

to be executed $n \times n \times 10$ times
 $\approx 10n^2$ So, the time complexity of func1 is $O(10n^2)$

• function 2: (void func2)

for (int i = 0; i < n*2; i++) {

for (int j = 0; j < i/3; j++) {

for (int k = 0; k < j*j; k++) {

count++ } }

innermost loop
goes from

$$0 \text{ to } j^2$$

Outermost loop

$$i \rightarrow 2n$$

$$0, 1, 2, \dots, 2n$$

$$0 + 1 + 2 + 3 + 4 + 5 + \dots + 2n = \frac{2n(2n+1)}{2}$$

inner loop

$$j = 0 \rightarrow j = i/3$$

$$i = 0$$

$$0$$

$$i = 5 \quad 0, 1$$

$$i = 1$$

$$0$$

$$i = 6 \quad 0, 1$$

$$i = 2$$

$$0$$

$$i = 7 \quad 0, 1, 2$$

$$i = 3$$

$$0$$

$$i = 4$$

$$0, 1$$

$$i = n \quad 0, 1, 2, \dots, \frac{2n}{3}$$

$$0 + 1 + 0 + 1 + 0 + 1 + 0 + 1 + 2 + 1 + 2$$

$$3 \times (1) + 3 \times (1 + 2) + 3 \times (1 + 2 + 3) + \dots$$

$$+ 3 \times (1 + 2 + 3 + \dots + \frac{2n}{3})$$

$$3 \left(1 + 1 + 2 + 1 + 2 + 3 + \dots + 1 + 2 + \dots + \frac{2n}{3} \right)$$

$$3 \left(1 + 4 + 6 + 10 + 15 + \dots + \frac{(2n+1)2n}{3} \right)$$

$$3 \left(\frac{(2n+3)(2n)}{2 \times 3} \right) = \frac{(2n)(n)}{3} = \frac{2n^2}{3}$$

innermost loop

$$K \geq 0 \text{ to } K < j^2$$

$$j=0$$

$$0$$

$$j=1$$

$$0$$

$$j=2$$

$$1, 2, 3$$

$$j=3$$

$$1, 2, 3, 4, 5, 6, 7, 8$$

$$j = \frac{2n}{3}$$

$$1, 2, 3, \dots, \frac{(2n)^2}{9}$$

$$(1+2+3) + (1+2+3+4+5+6+7+8) + \dots + (1+2+\dots+\frac{4n^2}{9})$$

$$\sum_{j=0}^n \frac{(j^2-1)(j^2)}{2} = \left(\frac{4n^2-1}{9}\right) \left(\frac{4n^2}{9}\right) \frac{1}{2}$$

ignoring constants

$$= \left(\frac{4n^2-1}{9}\right) \left(\frac{4n^2}{9}\right) \frac{1}{2} = \left(\frac{4n^2}{9}\right) \left(\frac{2n^2}{9}\right)$$

$$= \frac{8n^4}{81}$$

$$O\left(\frac{8n^4}{81} \times \frac{2n^2}{3} \times 2n^2\right) = O\left(\frac{16n^8}{243}\right)$$

Time Complexity

function 3 (Void func3).

```
for (int i=0; i<n; i++) {
    for (int j=i; j<n; j++) {
        for (int k=j; k<n; k++) {
            cout<<k<<" ";
        }
    }
}
```


Outermost loop

int $i=0$ to $i \leq n$.
n times.

~~$O(n)$~~

Inner loop

for $i=0$

0, 1, 2, ——— n

$i=1$

1, 2, 3 ——— n

$i=2$

2, 3, 4, ——— n

$i=n$

n

$$(0+1+2+ \dots + n) + (1+2+3+ \dots + n) + \dots + n$$

~~$\frac{n(n+1)}{2}$~~

~~$n \left(\frac{n(n+1)}{2} + \frac{(n-1)(n)}{2} + \dots + \frac{(n-(n-1)+1)(1)}{2} \right)$~~

~~$n \left(\frac{n(n+1)}{2} + \frac{(n-1)n}{2} + \frac{(n-2)(n-1)}{2} + \dots + \frac{(n-(n-1)+1)(1)}{2} \right)$~~

~~$n^2 + 2(n) + 2(n-1) + 3(n-2) + \dots + 0$~~

~~$n(n+1)$~~

$O(n^2)$

Innermost loop

for $j=0$

0, 1, 2, 3 ———

Similar case as above

shown

$O(n^2)$

Time complexity

$$n \times n^2 \times n^2 = n^5$$

~~$O(n^5)$~~

Ans

P. 7.0

(b) Total days calculated = N
 Array length = N

rain[] = {20, 15, 25, 18, 9, 30}

~~let Array length = N (given).~~

~~void BinarySearch (int arr[], int n,~~

~~void BinarySearch (int arr[], int n,~~

~~int i = 0;~~

~~int start = 0;~~

~~int size = n;~~

~~int end = arr[size-1];~~

~~int start~~

~~int mid = end + $\frac{(start - end)}{2}$;~~

~~if mid >~~

~~void FuturePredictor (int arr[], int n~~

~~int i = 0~~

~~int j = i + 1~~

~~for (int i = 0; i <= n; i++) {~~

~~if (arr[i] < arr[j]) {~~

~~j++;~~

~~for (int i = 0; i <= n; i++) {~~

~~for (int j = i + 1; j <~~

~~while (arr[i] < arr[j]) {~~

~~j++~~

~~int FuturePredictor (int arr[], int n~~
~~int j = j; int i = 0;~~
~~int j = i + 1;~~


```
for (i=0; i<=n-1; i++) {
```

```
  for (j=i+1; j<=n; j++) {
```

```
    if (arr[i] < arr[j]) {
```

~~BRUTE~~
~~FORCE~~

```
      break;
```

```
      count <= j-i << "days"
```

```
      break;
```

```
    } else return 0;
```

```
  }
```

```
return
```

arr = {20, 15, 25, 10, 9, 30}

~~but make it~~

```
for i=0
```

20 > 15

20 < 25 Break Count 25-20

= 5 days

①

i=2

15 < 25

15 < 18

15 > 9

Break Count 15-9

= 6 days

i=5

return 0.

Work can find complexity.

Outer loop → ~~outer~~ (n-1) times.

Inner loop → for i=0

(n-1) - (0+1)

(n-2) times

for i=1

(n-1) - (1+1)

(n-3) times

for i=(n-1)

(n-1) - (1+n-1)

$$\begin{aligned}
 & (n-2) + (n-3) + (n-4) + \dots + (n-(n+1)) \\
 & n \left(n - [1+2+3+4+\dots+(n+1)] - 1 \right) \\
 & n \left(n - \frac{(n+1)n}{2} - 1 \right) \\
 & n \left(\frac{2n - n^2 - n - 2}{2} \right) \\
 & \frac{2n^2 - n^3 - n^2 - 2}{2}
 \end{aligned}$$

for $i = (n-1)$ $(n-1) - (n-2) = 1$

$$(n-2) + (n-3) + \dots + (n-(n-1))$$

$$n \left(n - (2+3+\dots+n-1) \right)$$

$$n \left(n - (1+2+3+\dots+n-1) - 1 \right)$$

$$n \left(n - \frac{(n-1)n}{2} - 1 \right)$$

$$n^2 - \frac{n^2(n-1)}{2} - n$$

$$\frac{2n^2 - n^3 - n^2 - 2n}{2}$$

$$n(n-2) - (2+3+\dots+n-1)$$

$$n^2 - 2n - \left[(1+2+\dots+n-1) - 1 - n \right]$$

$$n^2 - 2n - \frac{n(n+1)}{2} - 1 - n$$

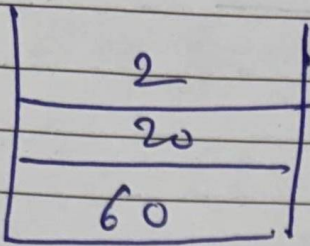
$$n^2 - 2n - \frac{n^2}{2} + \frac{n}{2} - 1 - n$$

$$\frac{n^2}{2}$$

Time Complexity

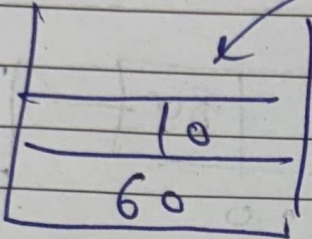
$$O\left(\frac{n^2}{2}\right)$$

Q: 2 (9) postfix $\rightarrow 60, 20, 2, 1, +, 15, 5, *, -, 100, +$



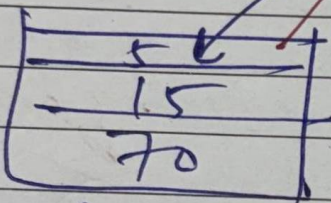
Stack

$$20 / 2 = 10$$



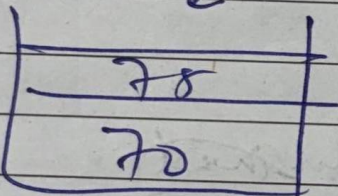
Stack

$$60 + 10 = 70$$



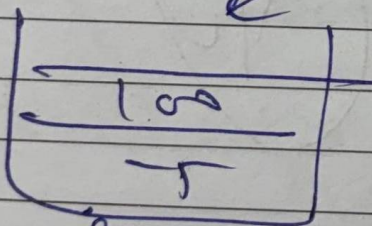
Stack

$$15 * 5 = 75$$



Stack

$$70 - 75 = -5$$



Stack

$$100 + (-5) = 95$$

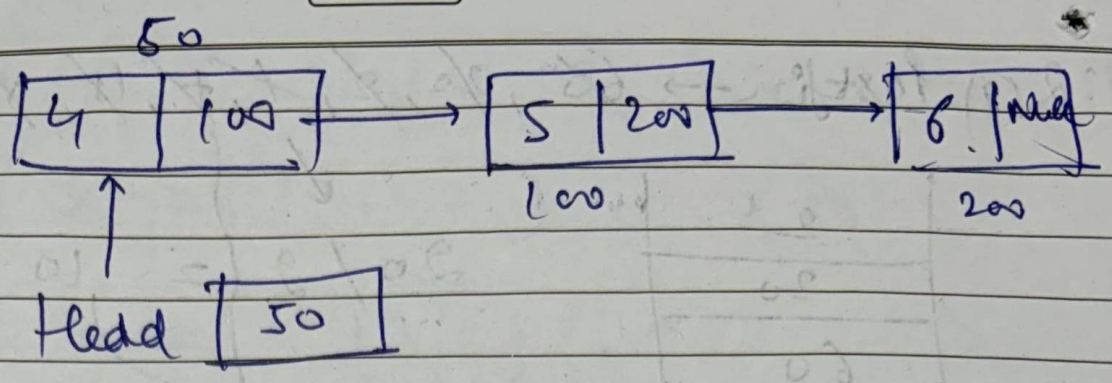
Ans = 95

(6)

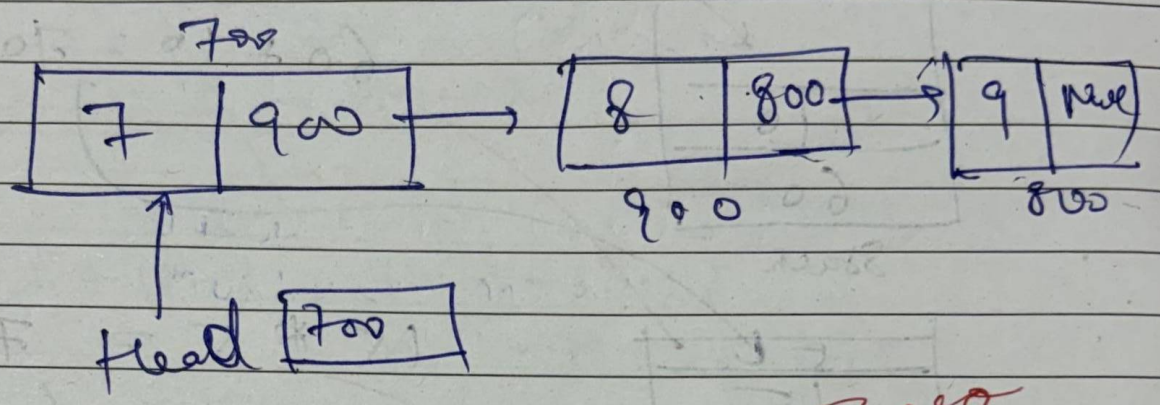
x

Q3: (a)

(U1)



(U2)



Zero

Null → Head * 700

delete → Head ?

Both linked lists combined.

VU

Q: 3(b) arr [0 — n-1]

Void BinarySearch (int arr [], int n, int x.)

```
int (i = 0; i < n; i++) {
    arr[i] > x //
    arr[i+1] < x;
    break;
    cout << i << endl; }
```

```
int n = size; int start = 0; //
int end = size - 1; //
int tar = arr[i];
int mid = end + (start - end) / 2;
```

~~if (tar < arr[mid]) {~~

~~end = mid; }~~

~~for (j = 0; j < n; j++) {~~

~~if (tar > arr[mid]) {~~

~~start = mid; }~~

~~else if (tar < arr[mid]) {~~

~~end = mid; }~~

~~else {~~

~~return mid~~

~~that value = mid;~~

value = arr[mid];

position = mid; }

TET

12/16

```
for (j=0; j<n; j++){
    int mid = end + ((start - end)/2)
```

```
    if (tar < arr[mid])
        end = mid; }
```

```
    else if (tar > arr[mid])
        start = mid; }
```

```
    else
        value = arr[mid];
        position = mid;
}
```

```
arr[position+1] = n;
```

```
cout << endl
```

```
for (a=0; a<n; a++){
```

```
    cout << a << endl
```

```
}
```


Q: 9 (9) Selection Sorting

Arr [] = { 45, 36, 78, 10, 12 }

let find 1st
Smallest no. call

Iteration-1

{ 45, 36, 78, 10, 12 }

{ 36, 45, 78, 10, 12 }

②

{ 36, 45, 78, 10, 12 }

{ 36, 45, 78, 10, 12 }

{ 10, 36, 45, 78, 12 }

{ 10, 36, 45, 78, 12 }

{ 10, 12, 36, 45, 78 } sorted.

Arr [] = { 45, 36, 78, 10, 12 }

ordered.

selected.

12 > 10 ✓

✓

{ 45, 36, 78, 10, 12 }

10, 12

{ 10, 12, 36, 45, 78 }

sorted

{ 45, 36, 78, 10, 12 }

{ 45, 36, 10, 12, 78 }

{ 45, 10, 12, 36, 78 }

No, selection sort is not always a stable algorithm because it can also give a error but bubble sort is the best and stable algorithm.

- (b)
1. $\text{curr} > \text{minNode}$ ~~ss~~ $\text{prev} > \text{minNode}$
 2. $\text{curr} < \text{minNode}$ ~~ss~~ $\text{prev} < \text{minNode}$
 3. New Node
 - 4.
 - 5.
 - 6.
 - 7.
 - 8.
 - 9.
 - 10.
- zero

Selection

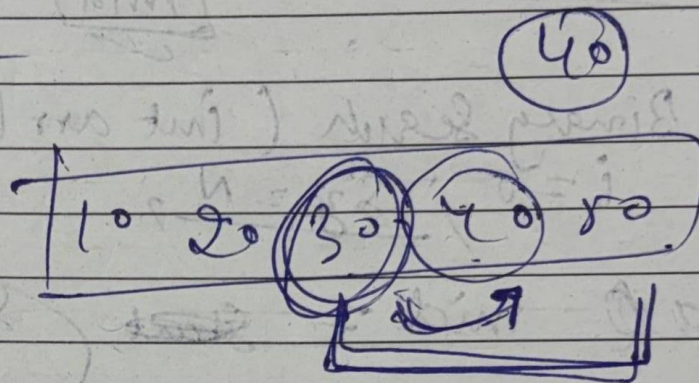
Insertion
15/16

Rough

(int arr []) {
n = size
i = 0
start = arr [i]
end = arr (size - 1)

int mid = end + start

~~select~~



atan mid